

Haoting (Alexa) Yu

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EDUCATION

University of British Columbia

BSc in Mathematics, Dean's Honor List

Relevant coursework: Calculus I-IV, Linear Algebra, Linear Programming, Partial differential equations, Probability, Stochastic Process

Vancouver, Canada

Sep 2018 - May 2022

University of Edinburgh

MSc in Operational Research with Data Science, Merit

Relevant coursework: Fundamental of Operational Research, Fundamental of Optimization, Operational Research, Applied Machine Learning, Reinforcement Learning, Biomedical Data Science

Edinburgh, UK

Sep 2022 – Sep 2023

University of the Arts London

MRes in Creative Computing

London, UK

Sep 2025 – Sep 2026

RESEARCH PROJECTS

From Retrieval to Responsibility: Designing Creative MRAG with Control and Governance [Poster] Accepted

- Publication: ACM Creativity & Cognition (C&C) 2026
- Research Focus: Research project on multimodal retrieval-augmented generation for creative practice. Developed a three-axis design framework for control and governance in creative MRAG, conducted semi-structured interviews with 10 visual and sonic creators, and derived design implications for provenance, interaction, and responsible reference-based generation.

Towards Robust Automatic Drum Transcription via Domain-Randomized Data Synthesis [Paper] Under Review

- Publication: ISMIR 2026
- Research Focus: Research project on synthetic data generation for Automatic Drum Transcription. Developed DITS, a domain-randomized synthetic drum dataset and procedural drum machine that randomizes rhythm, timbre, tempo, swing, effects, and instrument co-occurrence. Conducted ablation studies showing improved robustness on sonically diverse contemporary drum loops and competitive performance against real-data ADT baselines.

Who Chooses the Artwork? Curatorial Agency and Distributed Intent in Botto [Workshop Paper] Under Review

- Publication: Explainable AI for the Arts Workshop 2026, ACM Creativity & Cognition (C&C) 2026
- Research Focus: Research paper on authorship, explainability, and curatorial agency in AI art. Examined Botto as an agentic curatorial system where generation, ranking, voting, feedback, and recognition are recursively connected. Argued that authorship in Botto is distributed but asymmetrical, with creative agency shaped through selection, model filtering, DAO voting, and the process by which one generated image becomes formally recognized as the artwork.

PROJECT EXPERIENCE

Personalized Poem Generation with Fine-Tuned Large Language Model *Jan 2025 – present*

- Fine-tuned 7B LLM instruct model on 100 self-written poems using LoRA fine-tuning method to generate poetry that replicates my style, tone, and imagery, with title-based input for seamless poem generation

Anomaly Detection and Forecasting in Power Systems with Renewable Energy *June 2023 – Sep 2023*

- Developed machine learning models for time series anomaly detection and forecasting in power systems using seq2seq and autoencoder architectures, achieving over 90% detection accuracy and forecast deviations within 5%, contributing to more reliable grid operation amid growing renewable energy integration.

Development Plan of Electric Vehicle Charging Stations for the City of Dundee *Oct 2022 – Nov 2022*

- Designed a linear programming model using Xpress and Python to optimize the placement and scheduling of electric vehicle charging stations in Dundee, factoring in demand and potential charging locations

Sentiment Classification on Movie Reviews *Oct 2022 – Dec 2022*

- Developed a sentiment classification model for movie reviews using NLP techniques such as BERT, Bag of Words, and TF-IDF. Implemented logistic regression, SVM, and neural networks, optimizing model performance for a dataset of over 10,000 reviews

SKILLS

Language: Mandarin (native speaker), English (fluent)

Skills: Python, R, SQL, MATLAB, C++; PyTorch, TensorFlow, scikit learn, NumPy, Pandas, SciPy; prompt engineering, machine learning, multimodal AI, and generative AI prototyping with OpenClaw, Claude, Gemini, LangChain, and ChatGPT; data visualization and analytics using Power BI, Tableau, and Origin

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WORK EXPERIENCE

Fugro

Laboratory Quality Engineer

Wallingford, UK

Aug 2024 – Mar 2025

- Calibrate transducers and manage the Microsoft Access database, maintaining calibration records, test results, and equipment performance data to ensure measurement accuracy and streamlined lab operations
- Developed a full Python script to automate and replicate the lab's data processing Microsoft spreadsheet in Origin, improving efficiency, accuracy, and reproducibility of geotechnical data analysis
- Initiated an inter-lab comparison project using R, establishing a framework to evaluate and benchmark international lab performance, enhancing quality control and consistency across different locations
- Developed a baseline machine learning model in Python for soil particle and grain detection, utilizing YOLO v9 for image processing and object classification, enhancing automation in sediment analysis

Shanghai BaiLiYuan Pharmaceutical Group Co.

Data Analyst Intern

Shanghai, China

Dec 2023 - June 2024

- Conducted data analysis on pharmaceutical sales and distribution, leveraging Python (Pandas, NumPy) to analyse large datasets on product performance across various regions.
- Used SQL to query databases and extract relevant data on inventory levels, helping to optimize stock replenishment processes and reduce overstocking by 15%.
- Presented key findings through Power BI and Tableau to senior management, focusing on the correlation between marketing strategies and sales growth.

HD Education

Mathematics Science Instructor

Remote

May 2022 – Nov 2023

- Provided one-on-one and group tutoring sessions to undergraduate students, reinforcing concepts in subjects such as calculus, algebra, and statistics
- Developed tailored learning strategies to help students understand complex mathematical theories and problem-solving techniques.
- Assisted students in preparing for exams and improving their academic performance by offering clear explanations, practice problems, and constructive feedback.

UBC Faculty of Science

Math Department Teaching Assistant

Vancouver, Canada

Sep 2021 - May 2022

- Prepared course contents and weekly lab assignments
- Assisted students individually or in small groups with their assignments and related questions in facility workshop (teaching duty)

UBC Faculty of Graduate and Postdoctoral Studies

Doctoral Examinations Student Assistant

Vancouver, Canada

Sep 2021 - May 2022

- Provided daily administrative duties and support to the faculty, utilizing Microsoft Excel for data management and sorting examination-related information
- Collected, analyzed, and ensured data accuracy using Microsoft Excel for tracking and organizing data related to the doctoral examination process

Ipsos Strategy3

Consultant Intern

Beijing, China

April 2021 - Aug 2021

- Conducted desktop research and cold calls to gather data on the radiotherapy market in China, employing Microsoft Excel for data analysis
- Participated in in-depth interviews with industry experts, creating and updating the database using SQL for efficient data management and retrieval
- Collaborated with consultants to complete a project report for Varian, using Power BI and Microsoft PowerPoint to present market research insights and recommendations

Britbots Fund

Investment Analyst Intern

London, UK

May 2020 - Sep 2020

- Conducted research on the Chinese industrial robotics industry, performing DCF analysis, SWOT analysis, and P/S analysis on ZOA Robotics using Microsoft Excel and PowerPoint
- Researched various funding resources and utilized Power BI to create dashboards that highlighted the best investment opportunities for ZOA Robotics

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- Produced analytical reports with Tableau visualizations and used data modelling techniques in Python (Scikit-learn) to predict future market segmentation trends